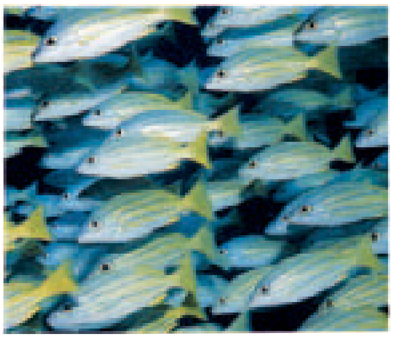
FLYING FISH
Gathering speed underwater, flying fish leap clear of the surface to escape predators, then glide for more than 30 seconds by spreading out the side fins.

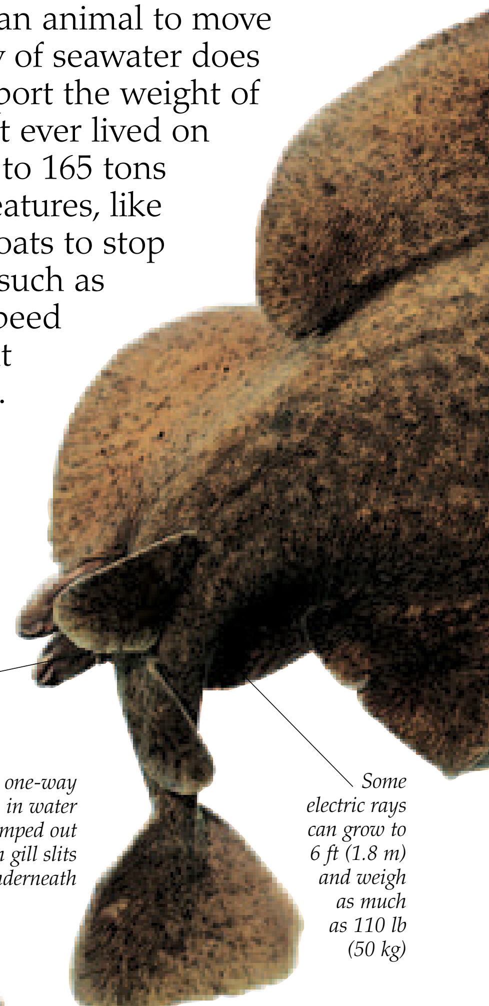


AT SCHOOL
Fish often swim together in a shoal or school (like these blue-striped snappers), where a single fish has less chance of being attacked by a predator than when swimming on its own. The moving mass of individuals may confuse a predator and also there are more pairs of eyes on the lookout for an attacker.

Moving along

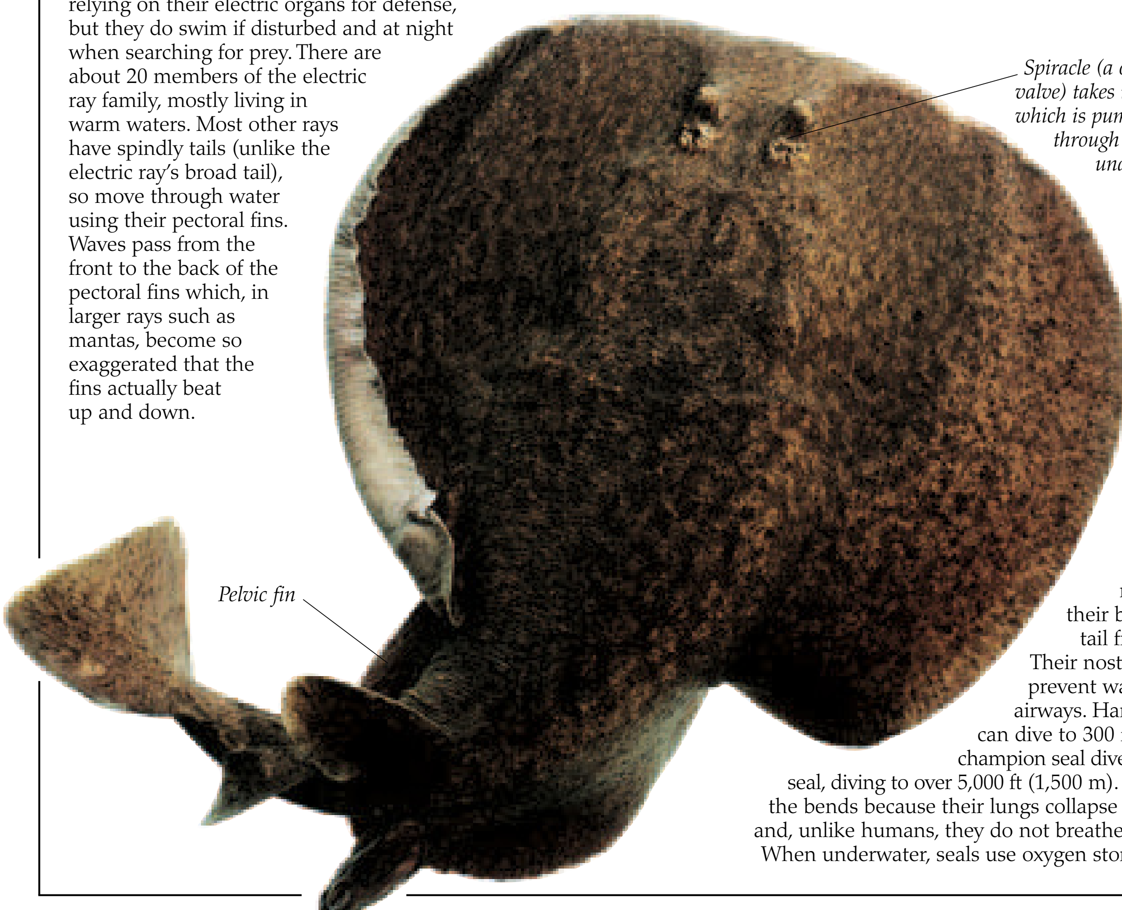
EVERY SWIMMER KNOWS that it is harder to move an arm or a leg through seawater than through air. This is because seawater is much denser than air. To be a fast swimmer like a dolphin, tuna, or sailfish, it helps to have a shape that is streamlined like a torpedo to reduce drag (resistance to water). A smooth skin and few projections from the body allow an animal to move through the water more easily. The density of seawater does have an advantage, in that it helps to support the weight of an animal's body. The heaviest animal that ever lived on Earth is the blue whale, which weighs up to 165 tons (150 metric tons). Some heavy-shelled creatures, like the chambered nautilus, have gas-filled floats to stop them from sinking. Some ocean animals, such as dolphins and flying fish, get up enough speed under water to leap briefly into the air, but not all ocean animals are good swimmers. Many can only swim slowly, some drift along in the currents, crawl along the bottom, burrow in the sand, or stay put, anchored to the seabed.

IN THE SWING
During the day, many electric rays prefer to stay hidden on the sandy bottom, as well as relying on their electric organs for defense, but they do swim if disturbed and at night when searching for prey. There are about 20 members of the electric ray family, mostly living in warm waters. Most other rays have spindly tails (unlike the electric ray's broad tail), so move through water using their pectoral fins. Waves pass from the front to the back of the pectoral fins which, in larger rays such as mantas, become so exaggerated that the fins actually beat up and down.



Electric ray's smooth skin can be either blackish or red-brown in color

Some electric rays can grow to 6 ft (1.8 m) and weigh as much as 110 lb (50 kg)



Spiracle (a one-way valve) takes in water which is pumped out through gill slits underneath

Pelvic fin

Swimming sequence of an electric ray

DIVING DEEP
True seals use front flippers to steer through the water. They move by beating their back flippers and tail from side to side. Their nostrils are closed to prevent water entering the airways. Harbor seals (right) can dive to 300 ft (90 m), but the champion seal diver is the elephant seal, diving to over 5,000 ft (1,500 m). Seals do not get the bends because their lungs collapse early in the dive and, unlike humans, they do not breathe compressed air. When underwater, seals use oxygen stored in the blood.